

Temperature-Calibrated Reservoirs: Safe Operating Regions and Test-Time Path Indicators

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Abstract

Reservoir computing relies on fixed recurrent dynamics, yet its standard hyperparameters mainly control recurrent scale, update timescale, hard sparsity, or reservoir size. We introduce *allocation temperature* as a distinct structural configuration axis. For each reservoir unit, it continuously sharpens or flattens the distribution of recurrent weight mass across incoming connections, while a separate normalization keeps the bulk recurrent scale comparable. Unlike hard sparsity, this control does not decide which allowed edges exist. We show that increasing allocation temperature monotonically increases row entropy and its effective support, defined as the entropy-equivalent number of equally weighted recurrent inputs.

Validation data select a default temperature and a connected region of nearby temperatures with similar validation performance. At test time, two label-free path-sensitivity indicators measure how much predictions change across this validation-safe region. Prediction dispersion summarizes disagreement across all safe temperatures, and endpoint spread compares the predictions at the low- and high-temperature ends of the region. These quantities indicate whether the model output is sensitive to the structural choice, but they do not identify which temperature will have the lowest target error.

Across controlled time-series tasks, standard reservoir benchmarks, and two public sensor datasets, allocation temperature produces a structural signature that differs from gain, leak rate, and hard sparsity. Validation-safe regions contain near-optimal test temperatures more often than same-width control intervals. This enrichment remains present with a denser 25-point temperature grid and with reservoirs of size $N = 150$, where the near-optimal containment rate is $2.16\text{--}2.86\times$ the rate of matched control intervals. Within the one-dimensional safe region, choosing the best safe temperature reduces the validation-selected default NRMSE by $0.18\text{--}1.91\%$ in the reference robustness panel, so the direct performance gain is modest. The path-sensitivity indicators remain positively associated with the improvement available inside the safe region, with correlations of $0.41\text{--}0.58$, although fixed binary thresholds are less stable. In the retained historical real-data summaries, Appliances Energy shows a null safe-region-versus-control comparison, whereas UCI Air Quality gives a positive result; the current public-data reconstruction notebooks recover the structural path but not these exact operational point estimates. A final distribution-shift study broadens the candidate set to an 11×11 temperature-leak grid. An upper-bound five-model ensemble optimized retrospectively with held-out target labels reduces NRMSE by $25.3\text{--}39.0\%$ relative to the best fixed structure chosen before deployment. A sparse ensemble fitted only on pre-deployment conditions recovers smaller gains of $12.0\text{--}19.9\%$ without held-out target labels, but degrades under block missingness. Overall, allocation temperature adds an interpretable structural dimension to reservoir design and exposes when recurrent allocation matters, while the experiments also delineate the limits of label-free test-time selection.

Keywords: reservoir computing; echo state networks; recurrent neural networks; structural regularization; distribution shift; sparse aggregation; safe operating regions; temperature softmax.

1 Introduction

Reservoir computing is attractive because it shifts most learning complexity from recurrent training to readout training. Echo state networks and liquid state machines use a fixed recurrent dynamical system to transform input histories into high-dimensional state trajectories, after which a linear or shallow readout can be trained efficiently (Jaeger, 2001; Maass et al., 2002; Lukosevicius and Jaeger, 2009). The recurrent substrate is therefore a central part of the model: it determines how input history is stored, mixed, filtered, and separated before the readout sees it.

Common design choices include spectral radius, input scaling, leak rate, sparsity, reservoir size, and random weight distribution (Schrauwen et al., 2007; Lukosevicius and Jaeger, 2009; Yildiz et al., 2012). These controls affect different properties. Spectral radius and gain primarily change recurrent amplification. Leak rate changes the state-update timescale. Hard sparsity changes the zero/nonzero connectivity pattern. This paper introduces a complementary structural question: how concentrated should the recurrent weight mass of each reservoir unit be across its incoming connections?

We summarize this concentration with the row’s *effective recurrent support*. The term does not count nonzero edges and does not make a binary decision about whether an edge is present. It is the entropy-equivalent number of equally weighted inputs, $S_{\text{eff}} = \exp(H)$. A sharply concentrated row has low effective support; a diffuse row has high effective support. Because the measure is continuous, values between integers represent graded concentration patterns. This is fundamentally different from hard top- k sparsity, which explicitly removes edges.

We control this graded allocation with a signed row-wise softmax. The parameter τ is called *allocation temperature* because temperature is also used elsewhere in machine learning to rescale output logits or token probabilities. Here it acts inside the recurrent weight construction, not at the model output. Each row has fixed raw scores a_{ij} and signs $s_{ij} \in \{-1, +1\}$, and allocates recurrent mass according to

$$p_{ij}(\tau) = \frac{\exp(a_{ij}/\tau)}{\sum_k \exp(a_{ik}/\tau)}, \quad W_{ij}(\tau) = c(\tau)s_{ij}p_{ij}(\tau), \quad (1)$$

where $c(\tau)$ is a global normalization factor chosen to keep the overall recurrent scale comparable across temperatures. This normalization is fixed by a target scale set before the temperature scan, so it is not re-selected separately for each τ . Low allocation temperature concentrates recurrent mass on a few high-score edges. High allocation temperature spreads it more evenly. Allocation temperature therefore defines a smooth structural path through recurrent connectivity while recurrent gain is controlled separately.

The first practical use of this path is conventional: validation can select one temperature as a fixed hyperparameter. The second use is diagnostic. Validation can identify a connected region of nearby temperatures whose performance remains close to the validation optimum. At test time, we can compare the predictions produced by the candidates inside this region. If they agree, changing allocation temperature cannot substantially change the output on that window. If they disagree, the output is sensitive to the structural choice. We call the resulting quantities *path-sensitivity indicators*. They require no test labels. They measure whether changing temperature could matter, not which temperature would be best.

This distinction between *sensitivity* and *target error* is important. Prediction disagreement can be observed without labels, whereas determining which candidate has the lowest target error requires information about the target. The final validation study therefore asks a broader question under distribution shift: how much improvement is available if target labels are allowed retrospectively to choose or combine reservoir structures, how much can be recovered by an ensemble fixed before deployment, and whether unlabeled target-window statistics can predict

which candidate will perform best.

This paper contributes:

1. A new reservoir configuration axis, allocation temperature, that continuously controls recurrent weight concentration while recurrent scale is normalized separately.
2. A theoretical characterization of this axis through row entropy and effective support, including monotonic increase of both quantities with temperature.
3. A validation procedure that selects a default temperature and a connected validation-safe region $\mathcal{T}_{\text{safe}}$.
4. Two label-free test-time path-sensitivity indicators that quantify prediction variation across $\mathcal{T}_{\text{safe}}$.
5. Controlled tests showing that validation-safe regions transfer to held-out data and outperform same-width control intervals.
6. Robustness experiments showing that safe-region enrichment and the continuous indicator–improvement relationship persist with a denser temperature grid and a larger reservoir.
7. Direct comparisons with gain, leak rate, and hard sparsity, plus two public sensor datasets, showing that allocation temperature is structurally distinct but not universally the best operational control.
8. A final distribution-shift study that compares three practically different quantities: the improvement available when target labels are used retrospectively, the performance of a sparse ensemble fitted only on pre-deployment conditions, and the ability of unlabeled target statistics to predict low-error candidates.

The paper separates these contributions deliberately. Allocation temperature is the main structural contribution. The safe region and path-sensitivity indicators characterize how that new axis behaves when calibrated from validation data. The final distribution-shift study then measures how much broader structural opportunity exists and how difficult it is to exploit that opportunity without target labels.

The remainder of the paper is organized as follows. Section 2 positions the method relative to reservoir design, adaptive recurrent models, and test-time adaptation. Section 3 develops the allocation-temperature construction and its entropy interpretation. Section 4 describes calibration, uncertainty estimation, and the distribution-shift study. Sections 5–9 provide the one-dimensional validation, comparison, real-data, and robustness results. Section 10 presents the final distribution-shift study. Sections 11–13 discuss interpretation, limitations, and conclusions. Appendix C provides the file-level reproducibility map for the reported experiments.

2 Related Work

Reservoir computing and echo state networks. Reservoir computing trains readouts over fixed recurrent dynamics (Jaeger, 2001; Maass et al., 2002; Lukosevicius and Jaeger, 2009). Much work has studied spectral radius, input scaling, leak rate, sparsity, topology, and state dimension. The present work controls a different property: the continuous concentration of recurrent weight mass within each row while keeping bulk recurrent scale comparable. Allocation temperature changes how sharply a row favors a small subset of incoming connections; a pure gain path multiplies a fixed recurrent shape, and hard sparsity changes the zero/nonzero edge pattern.

Reservoir structure, memory, and dynamics. Reservoir performance depends on a balance between memory, nonlinearity, separation, and stability (Yildiz et al., 2012). Hierarchical and multi-timescale reservoirs explicitly combine different dynamical regimes (Manneschi et al., 2021). Recent work has also used entropically regularized optimal transport to create dynamic, input-conditioned inter-assembly coupling in echo-state networks (Singh and Raman, 2026). That mechanism computes a doubly stochastic transport plan online and changes routing between reservoir assemblies. Allocation temperature is different: it indexes a static family of row-wise recurrent allocations generated from fixed scores and signs, and validation selects operating points on that family before deployment.

Adaptive continuous-time and hybrid reservoir models. A related line of work makes recurrent computation controllable through continuous-time dynamics, neural ordinary differential equations, and adaptive time constants (Chen et al., 2018; Hasani et al., 2021, 2022). Echo State Transformer models combine reservoir memory with attention and include learned softmax-based adaptive leak mechanisms (Bendi-Ouis and Hinaut, 2025). These approaches modify online state dynamics or learn state-dependent controls. Allocation temperature addresses a narrower structural question: the recurrent substrate is static for each candidate, and the exposed control axis changes the concentration of recurrent allocation rather than learning a time-varying hidden-state model.

Intrinsic plasticity, adaptive reservoirs, and physical substrates. Intrinsic plasticity changes reservoir node response statistics to improve computational usefulness (Schrauwen et al., 2008). Physical reservoir computing also studies controllable substrates and task-dependent operating points (Appeltant et al., 2011; Tanaka et al., 2019; van der Sande et al., 2017). Allocation-temperature calibration shares the idea that substrate parameters matter, but focuses on a validation-defined structural path and test-time prediction sensitivity rather than online recurrent plasticity.

Test-time adaptation and unlabeled model selection. Modern test-time adaptation often uses entropy or consistency to update model parameters at deployment time (Wang et al., 2021; Niu et al., 2022). Our path-sensitivity indicators are related in spirit because they are computed on target windows, but they do not update model parameters and do not estimate target error directly. They measure how much predictions vary across a set of structures that was already accepted by validation. Selecting the lowest-error model without target labels is a harder unlabeled risk-estimation problem. The final distribution-shift study evaluates that distinction explicitly inside a reservoir structural bank.

Softmax temperature, attention, and safe optimization. Temperature is widely used to control softmax sharpness in probability calibration, attention-like mechanisms, and differentiable categorical relaxations such as the concrete and Gumbel-softmax distributions (Guo et al., 2017; Maddison et al., 2017; Jang et al., 2017). Graph attention networks normalize learned neighbor scores with a row-wise softmax (Veličković et al., 2018), but their coefficients are input-dependent and trained end to end. Here the raw recurrent scores and signs are fixed, and allocation temperature indexes a family of static recurrent substrates rather than calibrating output probabilities or computing sample-dependent attention. The term “safe region” also echoes safe optimization and Bayesian optimization (Snoek et al., 2012; Sui et al., 2015), but here it denotes a validation-defined connected interval over a finite structural path, not a probabilistic constraint set optimized online.

3 Temperature-Softmax Reservoirs as Structural Paths

3.1 Entropy-regularized recurrent allocation

For a fixed reservoir row i , the raw score a_{ij} measures how strongly source unit j is preferred as an incoming recurrent contributor to target unit i . Temperature first defines an unsigned row-allocation distribution

$$p_{ij}(\tau) = \frac{\exp(a_{ij}/\tau)}{\sum_k \exp(a_{ik}/\tau)}, \quad \tau > 0, \quad (2)$$

over candidate recurrent inputs j . Thus $p_{ij}(\tau)$ is the fraction of row i 's unsigned recurrent mass assigned to source j before the sign and global scale are applied. Given a fixed sign matrix $S \in \{-1, +1\}^{N \times N}$, define the signed allocation matrix

$$M_{ij}(\tau) = s_{ij}p_{ij}(\tau). \quad (3)$$

The final recurrent matrix is

$$W(\tau) = c(\tau)M(\tau), \quad (4)$$

where $c(\tau) > 0$ is a scalar chosen after the signed allocation matrix has been constructed. In the experiments we use

$$c(\tau) = \frac{s_0}{\sqrt{N} \text{RMS}(M(\tau))}, \quad (5)$$

where $\text{RMS}(M) = \sqrt{N^{-2} \sum_{ij} M_{ij}^2}$ is the element-wise RMS. This convention makes the element-wise RMS of $W(\tau)$ equal to $s_0/\sqrt{N}$, giving a comparable bulk recurrent scale across temperatures and reservoir sizes. The target scale s_0 is fixed before scanning temperature; in all reported temperature-path experiments we set $s_0 = 1$ unless otherwise stated. In principle, s_0 could be selected by validation like an ordinary recurrent-gain hyperparameter, but it is not re-selected across the temperature grid. This separation is central: $p_i(\tau)$ controls row-wise allocation entropy and effective support, while $c(\tau)$ keeps the overall recurrent scale comparable across temperatures. The separate gain-path baseline deliberately varies this global scale to test whether temperature effects reduce to ordinary gain effects.

The row-softmax in Eq. (2) is the standard solution of an entropy-regularized linear allocation problem:

$$p_i(\tau) = \arg \max_{p \in \Delta} \left[\sum_j p_j a_{ij} + \tau H(p) \right], \quad (6)$$

where $H(p) = -\sum_j p_j \log p_j$ and $\Delta = \{p : p_j \geq 0, \sum_j p_j = 1\}$ is the probability simplex. The score term rewards allocating mass to high-score recurrent inputs; the entropy term rewards spreading mass across inputs. This is the standard Gibbs or maximum-entropy construction: temperature sets the strength of the entropy term relative to the score term (Jaynes, 1957). A short derivation is given in Appendix A. As $\tau \rightarrow 0$, the distribution concentrates on the largest raw score. As $\tau \rightarrow \infty$, it approaches the uniform distribution.

3.2 Monotonic effective support and concentration

Let $H_i(\tau) = -\sum_j p_{ij}(\tau) \log p_{ij}(\tau)$ denote the entropy of row i 's unsigned allocation. Writing $Z_i(\tau) = \sum_j \exp(a_{ij}/\tau)$, define the allocation-weighted mean raw score

$$\mu_i(\tau) \equiv \mathbb{E}_{J \sim p_i(\tau)}[a_{iJ}] = \sum_j p_{ij}(\tau) a_{ij}.$$

Here the scores a_{ij} are fixed numbers; only the weights $p_{ij}(\tau)$ assigned to them change with temperature. With this notation, the entropy can be expressed as

$$H_i(\tau) = \log Z_i(\tau) - \frac{\mu_i(\tau)}{\tau}. \quad (7)$$

Differentiating this expression yields

$$\frac{dH_i}{d\tau} = \frac{\text{Var}_{p_i(\tau)}(a_i)}{\tau^3} \geq 0, \quad (8)$$

where

$$\text{Var}_{p_i(\tau)}(a_i) = \sum_j p_{ij}(\tau) (a_{ij} - \mu_i(\tau))^2.$$

Thus entropy increases monotonically with temperature, except in the degenerate case where all row scores are equal. Appendix A expands Eq. (7) directly from the softmax definition and then derives Eq. (8); the result assumes that the raw scores a_{ij} are fixed while τ varies.

For interpretability, we also report the effective support

$$S_{\text{eff},i}(\tau) = \exp(H_i(\tau)). \quad (9)$$

This quantity is the entropy-equivalent number of equally weighted recurrent inputs. It is a continuous measure of concentration, not a literal count of active or nonzero connections; it expresses the same entropy in units that are easier to interpret. If a row spreads mass uniformly over exactly k inputs, then $H_i = \log k$ and $S_{\text{eff},i} = k$. Equation 8 implies that $S_{\text{eff},i}(\tau)$ also increases monotonically with τ because the exponential is monotone. This result is independent of downstream task performance: increasing temperature has a structural effect before any readout is trained.

The monotonicity statement concerns the allocation distribution $p_i(\tau)$, not the post-normalized matrix scale. The scalar $c(\tau)$ in Eq. (5) affects the global magnitude of $W(\tau)$ but not the row entropy of $p_i(\tau)$. This is why temperature is separated from a simple gain path: changing gain multiplies a fixed recurrent shape, whereas changing temperature alters which recurrent inputs receive mass while the overall scale is controlled separately.

3.3 From structural control to empirical calibration

The monotonicity result identifies what temperature controls at the level of recurrent allocation: increasing τ makes each row distribution less concentrated and increases its effective support. It does not determine the consequences for the complete reservoir dynamics. Although the normalization $c(\tau)$ keeps the bulk recurrent scale comparable, redistributing recurrent mass changes the signed matrix $W(\tau)$ and can affect spectral radius, short-term amplification, contractivity and echo-state behavior, memory, state separation, nonlinear saturation, and ultimately task performance (Jaeger, 2001; Lukosevicius and Jaeger, 2009; Yildiz et al., 2012). Temperature should therefore be understood as a structural allocation axis that is distinct from, but not dynamically independent of, gain and leak-rate controls. We consequently do not infer a useful operating regime from allocation entropy alone. Instead, spectral-radius variation is audited as one possible dynamical confound, while validation performance identifies the candidate temperatures that remain empirically plausible operating points.

For a fixed input sequence u_t , reservoir dynamics are

$$x_{t+1}^{(\tau)} = \tanh \left(W(\tau) x_t^{(\tau)} + W_{\text{in}} u_t \right), \quad \hat{y}_t^{(\tau)} = r_\tau(x_t^{(\tau)}), \quad (10)$$

where r_τ is the trained readout for temperature τ . The reservoir state is obtained by iterating this recurrence sequentially through the input series. In the experiments, the state is initialized

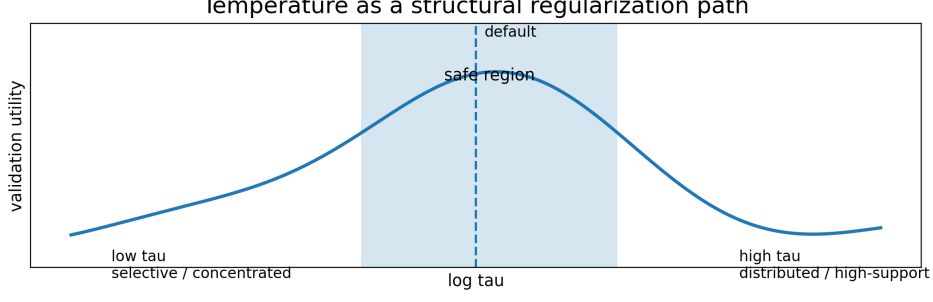


Figure 1: Conceptual view. Temperature defines a structural path over recurrent allocation. Validation selects a default temperature and a connected safe region around it. Test-time indicators measure whether predictions vary meaningfully inside this safe region.

at zero and an initial washout prefix is discarded before readout fitting or evaluation, reducing dependence on the arbitrary initial condition. No fixed-point solution is computed separately for each sample; $x_t^{(\tau)}$ is the state reached after processing the input history up to time t . Since $W(\tau)$ varies smoothly for $\tau > 0$, the state and prediction traces define a path over allocation temperature. This prediction path becomes operationally meaningful after validation identifies which temperatures remain plausible operating points for the task.

Let $U_{\text{val}}(\tau) = -\mathcal{L}_{\text{val}}(\tau)$ denote validation utility. We choose

$$\tau_{\text{default}} = \arg \max_{\tau \in \mathcal{T}} U_{\text{val}}(\tau), \quad (11)$$

where $\mathcal{T}$ is the discrete grid of tested temperatures. The validation-safe region is the connected component containing τ_{default} of the validation superlevel set

$$\mathcal{T}_{\text{safe}} = \{\tau \in \mathcal{T} : U_{\text{val}}(\tau) \geq U_{\text{val}}(\tau_{\text{default}}) - \epsilon\}. \quad (12)$$

The connected-component requirement prevents disjoint validation peaks from being merged into a single operating region. Thus the safe region is not a dynamical guarantee; it is an empirical, validation-defined subset of the structural path. Temperatures outside this set are not used for test-time movement.

3.4 Possible movement gain and path-sensitivity indicators

The best-safe gain on a labelled test window w is

$$G_{\text{safe}}(w) = \max_{\tau \in \mathcal{T}_{\text{safe}}} U_{\text{test},w}(\tau) - U_{\text{test},w}(\tau_{\text{default}}). \quad (13)$$

This quantity is not available at deployment because it requires labels. It is used only during validation to ask whether changing temperature inside the calibrated safe region could have helped on that window.

Two label-free path-sensitivity indicators measure whether predictions change enough across the safe region for temperature choice to matter. The first indicator is prediction dispersion across the validation-safe temperatures:

$$D_{\text{safe}}(w) = \frac{1}{|w|} \sum_{t \in w} \text{Std}_{\tau \in \mathcal{T}_{\text{safe}}} \left(\hat{y}_t^{(\tau)} \right), \quad (14)$$

where Std denotes the standard deviation across the predictions produced by the candidate temperatures in $\mathcal{T}_{\text{safe}}$ at the same time point. The second indicator is the low/high anchor spread,

$$A_{\text{LH}}(w) = \text{RMSE} \left(\hat{y}_w^{(\tau_{\text{low}})}, \hat{y}_w^{(\tau_{\text{high}})} \right), \quad (15)$$

where τ_{low} and τ_{high} are the endpoints of $\mathcal{T}_{\text{safe}}$. Low dispersion implies that all safe temperatures produce similar predictions, so changing temperature inside the safe region cannot substantially change the output. High dispersion does not guarantee improvement, but it shows that a labelled choice among the safe temperatures could change the prediction enough to matter.

The interpretation is direct. If all accepted operating points produce nearly identical predictions on a window, choosing among them cannot materially change the output. If the predictions differ, the model is sensitive to the structural choice on that window. This is why the indicators are evaluated as signals of possible improvement rather than as automatic model selectors.

4 Methods

The experimental design follows the same sequence used at deployment: construct comparable reservoir candidates, fit one readout per candidate, use validation data to select an operating region, evaluate that region on held-out data, and estimate uncertainty at the level of independent tasks, trials, splits, or reservoir substrates.

4.1 Constructing and evaluating the reservoir path

All one-dimensional temperature experiments use the signed row-wise softmax construction from Section 3. For a given reservoir instance, the raw scores, signs, input weights, and data split remain unchanged while allocation temperature varies. The normalization factor $c(\tau)$ is recomputed from Eq. (5) so that the bulk recurrent scale remains comparable across temperatures. With the element-wise RMS convention used here, $s_0 = 1$ gives recurrent entries with RMS $1/\sqrt{N}$ after normalization.

The robustness implementation makes the random substrate explicit. Off-diagonal raw scores are sampled independently from a standard normal distribution, recurrent signs are sampled independently with equal probability from $\{-1, +1\}$, self-connections are excluded, and input weights are sampled uniformly within the configured input-scale interval. The same raw-score matrix, sign matrix, and input matrix are then reused for every temperature in a given trial. The released experiment files specify the corresponding seeds and task-specific input scales.

Each candidate temperature defines a different recurrent matrix and therefore a different state trajectory. A separate ridge-regression readout is fitted for every candidate using the same training split, and all candidates are evaluated on the same validation and test splits. Reservoir states are generated sequentially from a zero initial state. An initial washout segment is discarded before fitting or scoring. Performance is measured by normalized root mean squared error (NRMSE), with normalization by the target standard deviation. Utility is defined as negative NRMSE so that larger utility is better.

An NRMSE improvement of 0.005 means that RMSE decreases by 0.5% of one target standard deviation. We therefore report both the gain available inside the validation-safe region and the corresponding absolute NRMSE values where they are available. This distinguishes modest improvements obtained by choosing among already accepted temperatures from the larger upper bounds obtained by searching or combining a broader structural bank.

4.2 Selecting a validation-safe region and computing test-time indicators

Training and validation determine the operating choices that are available before test targets are known. The finite set $\mathcal{A} = \{a_1, \dots, a_K\}$ denotes the candidate values along a one-dimensional reservoir path. For allocation temperature, $a_k = \tau_k$. The same calibration procedure is also applied to three comparison paths: recurrent gain, leak rate, and hard top- k sparsity. The

gain path multiplies a fixed recurrent matrix. The leak-rate path uses $x_{t+1} = (1 - \alpha_k)x_t + \alpha_k \tanh(Wx_t + W_{\text{in}}u_t)$. The hard-sparsity path retains the top- k scored recurrent inputs per row before scale normalization.

The absolute tolerance ϵ in Eq. (12) is obtained from a relative tolerance by

$$\epsilon = \max \left\{ \epsilon_{\text{abs}}, \epsilon_{\text{rel}} \left[\max_{a \in \mathcal{A}} U_{\text{val}}(a) - \min_{a \in \mathcal{A}} U_{\text{val}}(a) \right] \right\}. \quad (16)$$

Unless otherwise stated, $\epsilon_{\text{rel}} = 0.10$. The absolute floor follows the frozen protocol of each study: $\epsilon_{\text{abs}} = 0.005$ in the controlled synthetic/path experiments and $\epsilon_{\text{abs}} = 0.002$ in the two real-world sensor studies. The relative term scales the tolerance with the observed validation range, while the absolute floor avoids an unrealistically narrow threshold on nearly flat validation curves. Each study uses the same resulting numerical tolerance retrospectively when identifying near-optimal test candidates. In tolerance-sensitivity analyses, the study-specific ϵ_{abs} remains fixed and ϵ_{rel} is varied.

Algorithm 1: Calibration and test-time path-sensitivity workflow

Input: a reservoir path $\{W(a) : a \in \mathcal{A}\}$, training and validation data, a test stream, and tolerance ϵ .

1. Train one readout for every candidate operating point using the training data.
2. Measure validation performance at every candidate point.
3. Choose the validation-best point a_{default} as the default operating point.
4. Retain the connected neighboring candidates whose validation utility is within ϵ of the default. This is the validation-safe region $\mathcal{A}_{\text{safe}}$.
5. Record the low and high endpoints of the safe region.
6. On each test window, compare predictions across the safe candidates. Compute prediction dispersion and low/high endpoint spread.
7. Use these quantities to report how sensitive the prediction is to structural choices that were already accepted by validation. Target labels are required to determine which candidate has the lowest test error.

For retrospective evaluation, we identify the best test candidate on the tested grid and a near-optimal set consisting of candidates within the fixed utility tolerance of that best point. These labels are used only to evaluate whether validation selected a useful part of the path.

4.3 Validation questions

The experiments address five validation questions, each developed in a dedicated results section.

1. **Does the validation-safe region transfer to held-out data?** Section 5 measures exact and near-optimal containment, improvement over the validation-selected default, and comparisons with same-width control intervals. Table 2 and Fig. 2 summarize the main results.
2. **Do prediction differences across safe temperatures identify windows with more available improvement?** Section 5 relates prediction dispersion and endpoint spread to the improvement that could be obtained with labels, and then evaluates the same idea on standard reservoir tasks. Figure 3 and Fig. 4 show these results.

3. **Is allocation temperature distinct from established reservoir controls, and does the behavior transfer to measured sensor data?** Section 6 compares allocation temperature with gain, leak rate, and hard sparsity. Section 7 evaluates Appliances Energy Prediction and UCI Air Quality (Candanedo et al., 2017; De Vito et al., 2008). Tables 3 and 4 and Fig. 5 contain the corresponding results.
4. **Are the conclusions sensitive to the safe-region tolerance, spectral-radius changes, grid density, or reservoir size?** Section 8 examines safe-region width, tolerance, and structural confounds. Section 9 independently changes temperature-grid density and reservoir size. Tables 5–7 report the numerical results.
5. **How much structural improvement is available under distribution shift, and how much can be recovered without target labels?** Section 10 compares a fixed structure chosen before deployment, a sparse ensemble fitted on pre-deployment conditions, an unlabeled error-prediction model, and an upper-bound ensemble fitted retrospectively with target labels. Tables 8 and 9 summarize the outcomes.

Appendix C provides a compact mapping from each results section, table, and figure to the corresponding computational notebook and released artifacts.

4.4 How uncertainty is estimated

We report 95% bootstrap confidence intervals around mean gains and other summary statistics. For each analysis, 30,000 bootstrap samples are generated, the statistic is recomputed for each sample, and the 2.5th and 97.5th percentiles form the interval.

The resampling unit matches the independent unit of the experiment. When an analysis produces one result per independent task, trial, or split, those units are resampled directly. When many overlapping test windows come from the same task and reservoir trial, all windows from a task-trial run are kept together and complete runs are resampled. This cluster bootstrap preserves the dependence among windows that share the same reservoir instance and test sequence.

The grid-and-size robustness study contains four tasks and four trials per task, giving up to 16 task-trial clusters for the continuous indicator analysis. Some clusters are omitted from that specific analysis when the safe region contains only one candidate because prediction variation across a single candidate is undefined. In the distribution-shift study, each task has three independent reservoir substrates. We therefore report the observed task means in Table 8 and avoid emphasizing task-specific resampled confidence intervals. For the comparison across shift types, the resampling unit is the task-substrate pair, giving nine independent clusters; all windows from a selected cluster are kept together. The reported percentile intervals summarize uncertainty across this finite set of clusters.

4.5 Distribution-shift study

The final validation study asks whether the structural degrees of freedom revealed by allocation temperature remain useful when the test distribution changes. It deliberately broadens the structural candidate set from the one-dimensional temperature path to a two-dimensional temperature-leak grid. This makes it possible to separate two questions: how much improvement is available in principle, and how much of that improvement can be obtained without target labels. The corresponding results are reported in Section 10.

Candidate bank and forecasting tasks. The study uses NARMA10, Mackey–Glass, and Lorenz- x . Each task is evaluated with three independent reservoir substrates of size $N = 45$.

For each task and substrate, the candidate bank is

$$\Theta_{\text{global}} = \{\tau_1, \dots, \tau_{11}\} \times \{\alpha_1, \dots, \alpha_{11}\}, \quad (17)$$

where allocation temperature is logarithmically spaced from 0.08 to 3.50 and leak rate is linearly spaced from 0.10 to 1.00. The 121 candidates share the same raw-score and sign substrate. Each candidate has its own ridge readout trained on the clean training sequence.

Conditions available before deployment. The policies that do not use target labels are fitted using a set of pre-deployment conditions. These include clean operation, two levels of additive AR(1) input noise, and two levels of multiplicative gain drift. AR(1) noise is temporally correlated input noise rather than independent white noise. Gain drift changes the input amplitude multiplicatively. Six episodes are generated for each of these conditions.

Held-out test shifts. Evaluation uses shift conditions that are not used to fit the deployable policies: stronger additive AR(1) noise, two levels of additive offset drift, and two levels of block missingness. Block missingness removes contiguous portions of the input rather than merely perturbing their values. Four episodes are generated for each held-out condition. Every episode contains a 180-step context segment followed by a 180-step segment on which NRMSE is scored after washout. Target values from these held-out segments are used only for retrospective evaluation and for the explicitly label-informed upper bound described below.

Policies compared. Four alternatives are evaluated within the same candidate set.

1. **Fixed structure chosen before deployment.** One temperature–leak candidate is selected by its mean error across the pre-deployment conditions. It is then used unchanged on held-out shifts.
2. **Sparse ensemble fitted before deployment.** Up to five candidate predictions are combined with nonnegative weights that sum to one. The weights are fitted only on the pre-deployment conditions:

$$\hat{y}_w^{\text{pre}} = \sum_{\theta \in \Theta_q} w_{\theta}^{\text{pre}} \hat{y}_w^{(\theta)}, \quad w_{\theta}^{\text{pre}} \geq 0, \quad \sum_{\theta} w_{\theta}^{\text{pre}} = 1, \quad \|w^{\text{pre}}\|_0 \leq 5, \quad (18)$$

where $q \in \{\text{global}, \text{cert}\}$ identifies the candidate set used in that comparison.

3. **Unlabeled error predictor.** A model fitted on pre-deployment episodes predicts the error of each structural candidate from target-free features. It combines a 250-tree ExtraTrees model with standardized ridge regression. Features describe the structural coordinates, clean-validation error, prediction level and variance, disagreement with the candidate consensus, response to a small input perturbation, and reservoir-state magnitude and shift. Grouped cross-validation on the pre-deployment episodes selects the ridge penalty and the shrinkage toward each candidate’s historical mean error. At test time, the three candidates with the lowest predicted error are averaged.
4. **Target-label upper bound.** For retrospective comparison only, the same nonnegative five-candidate ensemble class is fitted using the target values of each held-out window. This reports how much improvement would be available if the target errors were known. It is not used as a deployable policy.

Table 1: Experimental scope. Section references identify where each study is described and interpreted. The notebook mapping is provided separately in Appendix C.

Study	N	K	Paths	Data lengths	Results
Path comparison	60	13	temperature, gain, leak rate, sparsity	1200/500/500	Sec. 6
Appliances Energy	50	13	temperature, gain, leak rate, sparsity	2500/1000/1000	Sec. 7
UCI Air Quality	50	13	temperature, gain, leak rate, sparsity	1500/600/600	Sec. 7
Grid and size robustness	60/150	13/25	temperature	1200/500/500	Sec. 9
Distribution-shift study	45	121	temperature $\times$ leak	1100/400; 180-step windows	Sec. 10

Pre-deployment certified subset. The study also evaluates a conservative subset of the 121 structures. Let $L_{\text{clean, val}}(\theta)$ be clean validation NRMSE. The initial subset contains candidates satisfying

$$L_{\text{clean, val}}(\theta) \leq 1.15 \min_{\theta' \in \Theta_{\text{global}}} L_{\text{clean, val}}(\theta'). \quad (19)$$

If fewer than 25 candidates satisfy this rule, the 25 candidates with lowest clean validation loss are retained. We denote the resulting set by Θ_{cert} . It is different from the connected one-dimensional temperature region $\mathcal{T}_{\text{safe}}$ in Eq. (12). Every comparison uses the same candidate set on both sides: methods evaluated on Θ_{global} are compared with an upper bound on Θ_{global} , and methods evaluated on Θ_{cert} are compared with an upper bound restricted to Θ_{cert} .

4.6 Experimental scope and computational cost

Table 1 summarizes the main experimental configurations. The one-dimensional comparison and real-data studies use 13 candidate values and reservoirs with 50 or 60 units. The robustness study separately increases the temperature grid to 25 points and the reservoir size to 150 units. The distribution-shift study uses 121 temperature–leak candidates with $N = 45$.

For a dense $N \times N$ recurrent matrix, one state update requires a matrix–vector product of order $O(N^2)$. Repeating that update for T time steps and K structural candidates gives $O(KTN^2)$ reservoir-simulation cost. For ridge regression with a design matrix of T rows and D readout features, forming $X^\top X$ costs $O(TD^2)$ and solving the resulting $D \times D$ linear system costs $O(D^3)$. Fitting one readout per candidate therefore costs $O(K(TD^2 + D^3))$, with $D \approx N + 1$ for a linear readout with bias. These expressions describe the dense implementation used for the complexity discussion; sparse reservoir matrices can reduce the recurrent simulation cost.

5 Core Validation Results

The core validation asks three simple questions. First, does the region selected from validation data contain useful temperatures on unseen test data? Second, is that success better than would be expected from an arbitrary interval of the same width? Third, can prediction differences inside the safe region identify test windows where changing temperature has more potential value? A final standard-task panel shows when these questions are informative and when the temperature path is effectively flat.

Table 2: Summary of the core validation questions. The core transfer experiment and the matched-control experiment use independently generated run panels and different denominators, so the two containment summaries are independent estimates. The two control rows evaluate different comparisons: validation-selected intervals versus same-width control intervals, and random movement inside versus outside the safe region.

Question	Evidence	Main result
Does the safe region transfer?	Best safe temperature versus the validation-selected default	Mean gain 0.00470 [0.00357, 0.00579]
Is the region better than an arbitrary interval?	Validation-derived region versus same-width control intervals	Mean interval-level gain advantage 0.02739 [0.01870, 0.03710]
Is movement safer inside the region?	Random movement inside versus random movement outside	Mean movement-gain advantage 0.03209 [0.02342, 0.04143]
Do prediction differences indicate opportunity?	High- versus low-indicator test windows	Mean gain difference 0.01697 [0.01228, 0.02197]
Does the behaviour appear on standard tasks?	Lorenz- x , Mackey–Glass, delayed memory, and NARMA10	Mean safe-region gain 0.00463 [0.00016, 0.00911]

5.1 Does the validation-safe region transfer to test data?

The first experiment asks whether validation identifies a useful neighborhood of temperatures around the selected default. Table 2 summarizes the numerical results and Fig. 2 shows the transfer and matched-control comparisons. On held-out test data, the safe region contained at least one near-optimal temperature in 78.1% of runs and contained the exact best temperature on the tested grid in 57.3% of runs. The best temperature available inside the safe region improved over the fixed validation default by a mean of 0.004702 NRMSE, with confidence interval [0.003565, 0.005786].

Because NRMSE is normalized by the target standard deviation, this mean gain is equivalent to reducing RMSE by about 0.47% of one target standard deviation. The best temperature over the entire grid produced a larger gain of 0.030254, or about 3.03% of one target standard deviation. The safe region is deliberately conservative: it captures the validation-supported part of the improvement available over the full path.

The next experiment measures whether the validation-selected region contains useful test temperatures more often than alternative intervals of exactly the same width. The core transfer experiment and the matched-control experiment use independently generated run panels and have different denominators. They answer related but distinct questions. The first estimates how often the validation-safe region transfers successfully. The second compares the validation-selected region with alternative intervals of exactly the same width. In that matched-control experiment, the validation-selected region contained a near-optimal test temperature in 79.2% of runs, compared with 38.5% for same-width control intervals. Exact-best containment was 63.5% for the validation-selected region and 15.6% for the controls. The mean gain advantage of the validation-selected region over the control intervals was 0.02739, with confidence interval [0.01870, 0.03710]. A separate movement comparison sampled candidate changes inside and outside the safe region. Moving to a candidate inside the region had a mean gain advantage of 0.03209 over movement outside it, with confidence interval [0.02342, 0.04143]. Figure 2 visualizes both comparisons. Together, these results show that validation identifies a non-arbitrary part of the allocation-temperature path.

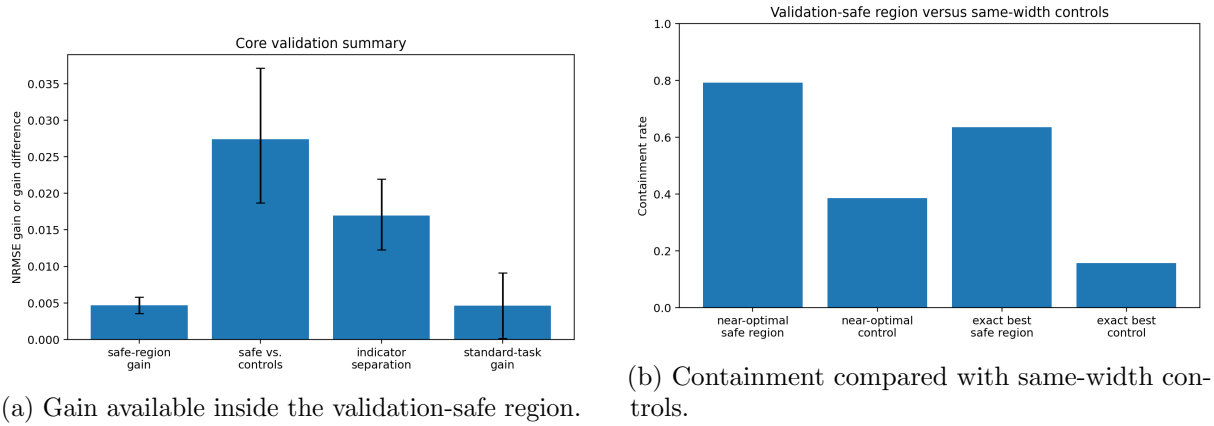


Figure 2: Core transfer and matched-control results. Panel (a) shows the improvement available inside the validation-safe region. Panel (b) compares the validation-selected region with same-width control intervals. The two experiments use independently generated run panels, so their containment percentages are not paired estimates.

5.2 Do prediction differences identify windows with more available improvement?

The path-sensitivity indicators are useful only if they are related to the improvement that target labels reveal retrospectively. Figure 3 reports this relationship. The simplest expectation is that, when all safe temperatures produce almost identical predictions, little can be gained by choosing among them. When their predictions differ more strongly, there is more room for temperature choice to affect the result.

This expectation is supported, although the relationships are moderate rather than deterministic. Across test windows, prediction dispersion was positively correlated with the improvement available from choosing the best safe temperature ($r = 0.412$). The low/high endpoint spread showed a similar correlation ($r = 0.405$). Test windows in the highest quartile of prediction dispersion had, on average, 0.01689 more available gain than windows in the lowest quartile, with confidence interval [0.01227, 0.02196]. The corresponding difference for low/high anchor spread was 0.01702, with confidence interval [0.01243, 0.02207].

These results show that stronger disagreement among safe temperatures tends to occur on windows where temperature choice matters more. The relationship is moderate rather than deterministic, so the indicators are used to quantify sensitivity rather than to select a direction automatically.

5.3 When is the temperature path informative?

The standard-task study tests how the same calibration behaves across Lorenz- x , Mackey-Glass, delayed memory, and NARMA10. Figure 4 shows the task-level pattern. It includes Lorenz- x , Mackey-Glass, delayed memory, and NARMA10. Across the panel, the safe region contained a near-optimal test temperature in 96.9% of runs and the exact best temperature on the tested grid in 78.1%. Mean safe-region gain was positive on all four tasks and averaged 0.004634, with confidence interval [0.000160, 0.009109]. This corresponds to about 0.46% of one target standard deviation in RMSE.

The task-level pattern is more informative than the average. Lorenz- x and Mackey-Glass have broad performance plateaus: the validation-selected default is already close to the best temperature, so little improvement remains inside or outside the safe region. Delayed memory

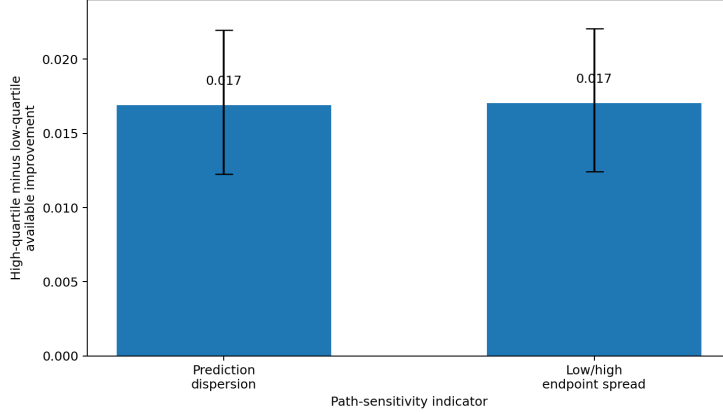


Figure 3: Windows with high prediction dispersion or high low/high endpoint spread have more available safe-region improvement than low-indicator windows. Bars show the high-quartile minus low-quartile difference in available improvement, with 95% bootstrap confidence intervals.

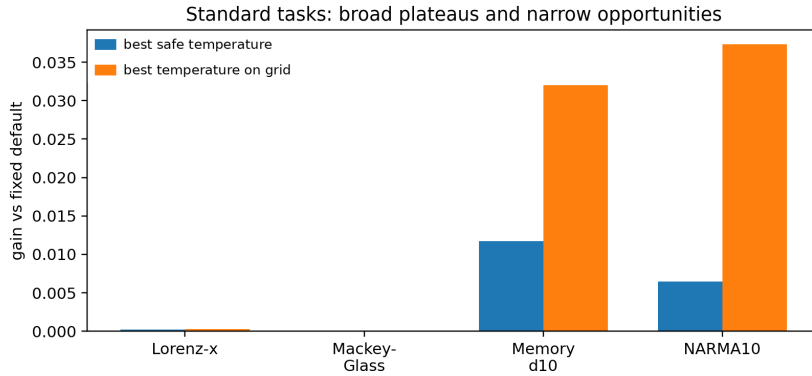


Figure 4: The value of allocation-temperature movement depends on the task. Lorenz- x and Mackey–Glass show broad plateaus with little additional improvement available over the tested temperature range, whereas delayed memory and NARMA10 show stronger sensitivity.

and NARMA10 are more temperature-sensitive and leave more room for movement along the full path. The path-sensitivity indicators are therefore most informative when allocation temperature produces meaningful variation in predictions and performance. On a broad performance plateau, little improvement is available for any indicator to reveal. The 78.1% exact-containment value in this separate four-task study happens to equal the near-optimal containment percentage of the earlier transfer experiment; the two values come from different panels and measure different events.

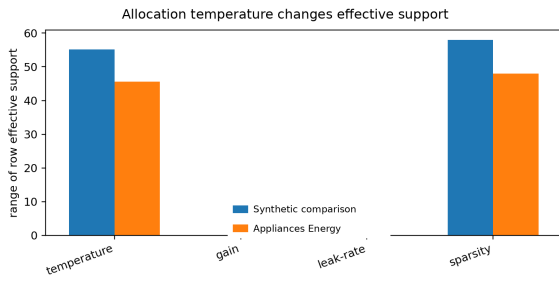
6 Comparison with Gain, Leak Rate, and Hard Sparsity

A key question is whether allocation temperature adds a genuinely different reservoir configuration axis or simply reproduces the effect of an established hyperparameter. Table 3 and Fig. 5 compare allocation temperature with recurrent gain, leak rate, and explicit top- k sparsity under the same calibration procedure.

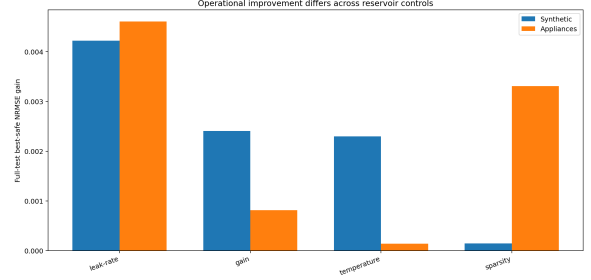
The comparison supports three conclusions. First, allocation temperature changes the concentration of recurrent weight mass in a way that gain and leak rate do not. Its mean effective-support range is 55.19 equivalent inputs, while gain and leak-rate paths leave this concentration measure

Table 3: Comparison of four one-dimensional reservoir controls. Allocation temperature changes the continuous concentration of recurrent weight mass while the bulk scale is normalized. Gain changes recurrent scale, leak rate changes the update timescale, and top- k sparsity changes the hard connectivity pattern.

Path	Effective-support range	Spectral-radius range	Full-test best-safe gain	Safe-control gain	Interpretation
Temperature	55.19	0.046	0.00230	+0.02303	continuous concentration path
Gain	0.00	1.438	0.00241	+0.03173	amplification/spectral-radius path
Leak-rate	0.00	—	0.00422	+0.06486	update-timescale path
Sparsity	58.00	task-dependent	0.00015	+0.04160	hard top- k support path



(a) Support range by path.



(b) Improvement available inside each validation-safe region.

Figure 5: Path ablations. Allocation temperature and sparsity change recurrent support; gain and leak-rate do not. Panel (b) reports full-test best-safe NRMSE gain under a common metric across controls. Operationally, allocation temperature is useful and structurally distinct but not dominant over all one-dimensional paths.

essentially unchanged. At the same time, the spectral-radius range along the temperature path is only 0.046, whereas the gain path changes spectral radius by 1.438 while leaving effective support fixed. These complementary signatures distinguish allocation temperature from ordinary amplification. Second, the temperature safe region remains enriched relative to same-width controls: its mean safe-minus-control gain is +0.02303 with a positive confidence lower bound. Third, under the common full-test best-safe metric, gain and leak rate provide larger within-safe-region improvement in the synthetic comparison. Allocation temperature should therefore be treated as an additional structural coordinate, not as a replacement for existing controls.

Hard sparsity is the closest structural comparison because it also changes how many inputs matter. The difference is that top- k sparsity sets excluded edges exactly to zero, whereas allocation temperature continuously redistributes weight mass across the allowed edges. In this sense, allocation temperature provides a continuous concentration control that can move between highly focused and diffuse recurrent rows without changing the hard connectivity pattern. The hard-sparsity path has a similar effective-support range (58.00) but smaller full-test safe-region improvement (0.00015) than allocation temperature (0.00230) in this comparison.

7 Real-World Sensor Benchmarks

Table 4 applies the same calibration procedure to two public sensor time-series datasets. The first study uses the UCI Appliances Energy Prediction dataset (Candanedo et al., 2017). The task predicts $\log(1 + \text{Appliances}_{t+h})$ from current appliance use, indoor sensor variables, weather variables, and time encodings, with horizon $h = 6$ samples, approximately one hour in the original ten-minute sampling. The second study uses the UCI Air Quality dataset (De Vito et al., 2008). The task predicts future benzene concentration $\text{C6H6}(\text{GT})$ from current multivariate

Table 4: Retained historical real-world sensor benchmark summary. Appliances Energy yields a null safe-region-versus-shuffled result, whereas Air Quality gives a positive safe-region result. The current reconstruction notebooks recover the documented structural path but not these exact operational point estimates; fresh reconstruction outputs are reported separately in the repository.

Dataset	Rows/features	Target	Temp. support range	Temp. full-test best-safe gain	Temp. safe – control	Temp. path indicators
Appliances Energy	19729/32	log future appliances, $h = 6$	45.62	0.00014	-0.01199 $[-0.05331, 0.01532]$	0.510 / 0.522
Air Quality	9356/19	future C6H6, $h = 1$	45.55	0.01635	$+0.05312$ $[0.00695, 0.09534]$	0.727 / 0.646

chemical-sensor and meteorological measurements, with horizon one hour.

The retained historical summaries provide complementary evidence for how the allocation-temperature axis behaved on measured sensor data. On Appliances Energy, allocation temperature retains its structural signature and its two path-sensitivity indicators remain informative, but its full-test best-safe gain is only 0.00014, while the safe-region-minus-shuffled point estimate is negative (-0.01199) and its confidence interval crosses zero ($[-0.05331, 0.01532]$). This is a null result and does not support safe-region superiority on that dataset; leak-rate also provides stronger available safe-region improvement there. On Air Quality, the retained result is stronger: the temperature safe region outperforms same-width controls, prediction dispersion and low/high endpoint spread correlate strongly with the improvement available inside the safe region, and allocation temperature gives the largest within-safe-region improvement among the four tested paths. Because the original June 2026 M20b/M20c2 notebook binaries could not be recovered, Table 4 reports retained historical summaries. The repaired public-data reconstructions preserve the documented dataset dimensions and structural temperature path but do not numerically recover these operational point estimates; fresh outputs are reported separately under `results/reproduced/`.

8 Additional Validation of Sensitivity and Structural Confounds

This section tests whether the main conclusions depend on the width of the validation-safe region, the tolerance used to define it, or unintended changes in spectral radius. It also separates the continuous path-sensitivity result from a less stable binary threshold summary.

8.1 Safe-region width and tolerance sensitivity

The validation safe region is intentionally conservative. At the default relative tolerance $\epsilon_{\text{rel}} = 0.10$, using each study’s native absolute floor, the temperature safe region occupies only a small fraction of the 13-point candidate grid: mean widths are 3.23 candidate points in the synthetic path-comparison study, 1.75 candidate points on Appliances Energy, and 2.50 candidate points on Air Quality (Table 5). Relative to all same-width intervals on the same grid, near-optimal containment is enriched by factors of 3.12, 4.25, and 3.63, respectively. These factors should be interpreted as finite-grid empirical quantities, not as grid-resolution-invariant constants. Under the reported grids, the enrichment over same-width controls shows that the safe region identifies a more informative part of the candidate path than interval width alone would provide. For the fully regenerated synthetic path-comparison audit, sensitivity checks over $\epsilon_{\text{rel}} \in \{0.05, 0.10, 0.15, 0.20\}$ preserve the qualitative conclusion: increasing tolerance widens the safe component while same-width interval comparisons remain the appropriate null model. The real-data rows below report the retained historical native $\epsilon_{\text{rel}} = 0.10$ summaries.

Table 5: Safe-region width at $\epsilon_{\text{rel}} = 0.10$ using each study’s native absolute tolerance. The validation-selected regions occupy a small fraction of the tested grid and contain near-optimal test points more often than same-width control intervals. Full-test best-safe gain is reported with the same definition in all rows.

Study	ϵ_{abs}	Mean safe width	Width fraction	Near-optimal containment ratio	Full-test best-safe gain
Synthetic path comparison	0.005	3.23/13	24.8%	3.12×	0.00230
Appliances Energy	0.002	1.75/13	13.5%	4.25×	0.00014
Air Quality	0.002	2.50/13	19.2%	3.63×	0.01635

8.2 Reliability of the path-sensitivity indicators

The main result is continuous: windows with greater prediction dispersion or greater low/high endpoint spread tend to have more improvement available from choosing among the safe temperatures. A secondary analysis converted each indicator into a top-quartile binary flag and asked whether flagged windows were more likely to have positive gain. That threshold result is less stable when grid density or reservoir size changes. We therefore retain the continuous association as the primary result and report the binary threshold tables separately in Appendix B.

8.3 Separating allocation temperature from gain and hard sparsity

The structural audit separates support from gain. Tables 3 and 4 show that allocation temperature changes effective support substantially, whereas gain and leak-rate leave it approximately fixed. The complementary spectral-radius comparison points in the opposite direction. Gain changes spectral radius by about 1.4 in the synthetic comparison and both real-data studies, while allocation temperature changes it only mildly (0.046, 0.083, and 0.042). The two controls therefore affect different measurable properties: gain primarily changes recurrent amplification, whereas allocation temperature primarily changes the concentration of recurrent weight mass.

Hard sparsity is the closest structural comparator because it also changes how recurrent influence is distributed across inputs. The distinction is both structural and operational. Sparsity imposes hard top- k truncation, setting excluded edges to zero. Allocation temperature continuously redistributes recurrent weight mass over the same candidate row, preserving graded weights. In the path comparison, sparsity has support range 58.00 but full-test best-safe gain only 0.00015, whereas the allocation-temperature path has comparable support expansion and full-test best-safe gain 0.00230 (Table 3). On Air Quality, allocation temperature is also the strongest operational path. Allocation temperature therefore provides a continuous concentration control that is distinct from hard support selection.

9 Robustness to Grid Density and Reservoir Size

This robustness study changes one experimental factor at a time. It retains delayed memory, NARMA10, and Lorenz- x , and adds a controlled delayed-signal task with a non-predictive distractor channel. The controlled task has a known delayed target relationship, which makes it useful for checking whether the conclusions survive changes in temperature-grid density and reservoir size. Each task is evaluated over four trials with train/validation/test lengths 1200/500/500 and a 100-step washout. The baseline configuration is $N = 60, K = 13$. A nested $K = 25$ temperature grid changes only the grid density at $N = 60$ while reusing the same data and reservoir substrate. The $N = 150, K = 13$ configuration changes only reservoir size. For each configuration, the validation-safe region and all same-width control intervals are recomputed from that configuration’s validation results. Table 6 summarizes the containment and indicator

Table 6: Grid-density and reservoir-size robustness. Indicator correlations measure the relationship between test-window prediction variation and the improvement available from choosing the best temperature inside the validation-safe region. The secondary binary threshold analysis is reported in Appendix B.

Configuration	Safe width	Near-optimal containment ratio	Best-point containment ratio	Safe – control gain	Disp. corr.	Endpoint corr.
$N = 60, K = 13$	20.7%	2.864×	3.513×	+0.03397	0.452	0.410
$N = 60, K = 25$	17.5%	2.160×	5.014×	+0.02725	0.515	0.513
$N = 150, K = 13$	18.8%	2.257×	3.304×	+0.02043	0.577	0.566

Table 7: Absolute test-performance context for the reference robustness configuration ($N = 60, K = 13$), averaged over four fixed trials per task. Safe reduction is $(\text{NRMSE}_{\text{default}} - \text{NRMSE}_{\text{best safe}})/\text{NRMSE}_{\text{default}}$. The best-grid value is computed retrospectively and shows the remaining improvement available over the tested temperature grid.

Task	Default test NRMSE	Best-safe test NRMSE	Best-grid NRMSE	Safe reduction
Controlled delayed signal	0.92665	0.91713	0.90874	1.03%
Delayed memory	0.77276	0.77136	0.77069	0.18%
NARMA10	0.66285	0.65836	0.65836	0.68%
Lorenz- x	0.03269	0.03206	0.02254	1.91%

results, and Table 7 gives the absolute NRMSE values.

A safe region containing a single temperature still contributes to the containment and gain analyses. The two path-sensitivity indicators, however, quantify variation across multiple candidate predictions, so they require at least two distinct safe temperatures. Singleton cases are therefore omitted from the indicator-correlation analysis. The primary robustness question is whether the continuous relationship between prediction variation and available safe-region improvement persists. A separate binary top-quartile analysis is retained only as a secondary summary.

Table 7 makes the practical magnitude explicit. In the baseline robustness setting, selecting the best temperature inside the validation-safe component reduces default test NRMSE by 0.18%–1.91% across tasks. These are modest NRMSE reductions obtained by choosing among temperatures that validation had already accepted. The best candidate over the full tested grid is reported separately as a retrospective reference. The core robustness conclusion is stable. Mean safe-region width remains between 17.5% and 20.7% of the candidate grid. Near-optimal containment occurs more than twice as often in the validation-safe region as in same-width control intervals, and exact-best containment occurs more than three times as often. These ratios describe enrichment in containment frequency. The corresponding prediction-error changes are reported separately in Table 7. Temperature also continues to sweep a large structural range: the effective-support range is $0.919N$, $0.919N$, and $0.943N$ in the three configurations, where N is the number of reservoir units, while spectral-radius ranges remain 0.182, 0.189, and 0.138, respectively. Thus the safe-region enrichment persists beyond the original $K = 13, N = 60$ setting, while the numerical containment ratios remain empirical quantities for the tested grids and reservoirs.

The path-sensitivity result is more nuanced. Safe-region dispersion and anchor spread remain positively correlated with available safe-region improvement in every configuration, with correlations between 0.410 and 0.577. High-minus-low mean available improvement has a positive point estimate for both primary indicators in all configurations. For $N = 150$, however, only 11 of 16 cases have multi-point safe regions and the bootstrap interval includes zero; the larger-reservoir result is therefore directionally consistent with the smaller configurations but does not independently confirm a nonzero high-versus-low difference. The binary top-quartile statistic behaves differently. Its precision ratio is close to one in the baseline configuration and below one after

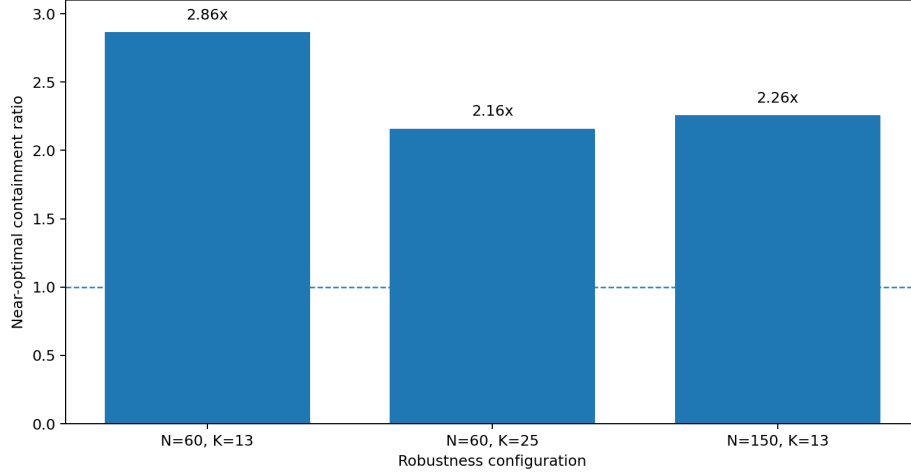


Figure 6: Near-optimal containment relative to same-width control intervals under the three robustness configurations. The validation-safe region remains enriched after increasing temperature-grid density or reservoir size. Continuous indicator correlations for the same configurations are reported in Table 6.

increasing grid density or reservoir size, with intervals crossing one. The stable result is therefore the continuous association: larger prediction variation along the calibrated temperature path tends to accompany more available improvement. The usefulness of a fixed binary threshold depends on the tested grid, reservoir size, and base rate of positive gain.

10 Structural Opportunity Under Distribution Shift

This is the final validation section. The earlier experiments ask how much can be gained by choosing among temperatures that remain inside a one-dimensional validation-safe region. Here the candidate set is deliberately broader: an 11×11 temperature-leak grid. The aim is to measure how much improvement is available under distribution shift when a second structural control is added, and then to test how much of that improvement can be recovered without target labels. The one-dimensional path indicators and this two-dimensional study therefore answer complementary questions. The former measure sensitivity inside an already accepted temperature region; the latter measures the larger structural opportunity and the difficulty of identifying low-error structures from unlabeled target data.

10.1 Upper-bound benefit of target-specific structural combinations

Table 8 compares three quantities on the full 121-candidate set. The fixed baseline is the single structure chosen from the pre-deployment conditions. The pre-deployment sparse ensemble can combine up to five structures, but its weights are also fixed before held-out evaluation. The third quantity is an upper bound: a five-model ensemble fitted retrospectively with the target values from each held-out window.

Table 8: Distribution-shift results by task. NRMSE values are averaged over held-out shift types and three independent reservoir substrates. Relative gains use $(L_{\text{fixed}} - L_{\text{policy}})/L_{\text{fixed}}$. Because only three substrates are available per task, the main table reports the observed task means rather than emphasizing resampled task-specific confidence intervals.

Task	Fixed	Pre-deployment ensemble	Target-label upper bound	Ensemble gain	Upper-bound gain
NARMA10	1.0064	0.8854	0.7518	12.02%	25.30%
Mackey–Glass	0.6193	0.4959	0.3776	19.93%	39.03%
Lorenz- x	0.6577	0.5274	0.4367	19.82%	33.60%

The target-label upper bound reduces NRMSE by 25.3%–39.0% relative to the fixed structure selected before deployment. Because each task contains only three independent reservoir substrates, the task means and their consistency across tasks carry the claim. We therefore do not emphasize task-specific resampled intervals in the main table. These reductions are much larger than the improvements available inside the one-dimensional validation-safe temperature region. The experiments are intentionally different: the upper bound can combine structures from a 121-candidate temperature–leak bank and uses target labels from each held-out window. It therefore measures the amount of structural improvement available in principle under shift.

The sparse ensemble fitted only on pre-deployment conditions reduces task-level NRMSE by 12.0%–19.9%. Relative to the target-label upper bound, it recovers approximately 48%–59% of the available reduction. A fixed combination of complementary structures can therefore recover a meaningful part of the shift-specific opportunity without held-out target labels, while a substantial gap remains.

10.2 Predicting low-error structures without target labels

The unlabeled error-prediction model does not close that gap. Its mean NRMSE is 0.9135 on NARMA10, 0.5403 on Mackey–Glass, and 0.5422 on Lorenz- x , compared with 0.8854, 0.4959, and 0.5274 for pre-deployment sparse ensemble. It improves on the fixed baseline at the task-average level, but is worse than the simpler pre-deployment sparse ensemble on all three tasks.

The error-prediction model uses structural coordinates, clean-validation and pre-deployment error, predictive disagreement and variance, perturbation consistency, and reservoir-state shift diagnostics. Even with these features, it does not transfer reliably enough to rank the candidate structures by held-out target error. This result reinforces the distinction developed earlier: prediction variation can reveal structural sensitivity, while estimating which structure will be most accurate is harder without target labels.

10.3 Performance across shift types and cost of pre-deployment certification

The average benefit of the pre-deployment sparse ensemble depends strongly on the type of held-out shift (Table 9). It is positive under stronger correlated additive noise and offset drift, but reverses under block missingness.

Table 9: Pre-deployment sparse ensemble by held-out shift type. Relative gains are computed against the pre-deployment fixed structure. Intervals resample the nine task–substrate clusters and retain all windows within each cluster. They remain finite-panel descriptive summaries. A negative gain denotes degradation.

Held-out shift	Pre-deployment fixed	Sparse ensemble	Relative gain [95% CI]	Windows harmed
Strong correlated additive noise	1.4606	0.9771	33.10% [23.08, 41.61]	0/36
Offset drift	0.6950	0.5932	14.65% [11.18, 16.82]	16/72
Block missingness	0.4776	0.5088	−6.55% [−15.11, −0.49]	45/72

Under block missingness, aggregation harms 45 of 72 held-out windows (62.5%), and the cluster-bootstrap interval remains below zero. The pre-deployment sparse ensemble is therefore beneficial on average across the task panel, but its reliability depends on the type of shift. The failure is especially important because block missingness removes information rather than only perturbing its scale or adding noise; a fixed combination of structural experts cannot restore observations that are absent.

Restricting the set of structures available to the target-label upper bound also has a measurable cost. We define the aggregate relative price of certification as

$$C_{\text{cert}} = \frac{L_{\text{cert}}^* - L_{\text{global}}^*}{L_{\text{global}}^*}, \quad (20)$$

Here L_{global}^* is the mean NRMSE of the target-label upper-bound ensemble when it can choose from all 121 structures. L_{cert}^* is the corresponding mean when the ensemble is restricted to the pre-deployment certified subset. Both use the same five-model nonnegative ensemble class. The cost is 4.98% for Lorenz- x (descriptive 95% interval 3.02–8.93%), 9.61% for Mackey–Glass (7.05–14.37%), and 8.29% for NARMA10 (1.86–16.76%). The certified subset therefore reduces the number of structures that can be considered at deployment, but it also excludes some combinations that perform well on particular held-out shifts. The percentages above quantify that performance trade-off.

11 Discussion

The main contribution is a new structural configuration axis for reservoir computing. Allocation temperature continuously controls how recurrent weight mass is concentrated across each row while recurrent scale is normalized separately. This provides a continuous control of recurrent weight concentration without changing the hard connectivity pattern, and a configuration dimension that is empirically distinct from gain and leak rate. The theoretical result establishes monotonic control of allocation entropy and effective support. The empirical program then characterizes how this axis behaves under validation, on measured sensor data, under changes in grid density and reservoir size, and under distribution shift.

The one-dimensional safe-region experiments show two complementary properties. First, validation selects temperature intervals that contain near-optimal test points more often than same-width control intervals. Second, test-window prediction variation across those accepted temperatures is positively associated with the improvement that target labels reveal retrospectively. The direct NRMSE reduction available inside the safe region is modest in the reference

robustness panel (0.18%–1.91% from the validation-selected default). The main value of these indicators is therefore diagnostic: they show when recurrent allocation choice matters on the current window. The continuous relationship persists when the temperature grid is made denser and when the reservoir grows to $N = 150$. The binary top-quartile rule is less stable and is retained only as a secondary analysis in Appendix B.

The path comparison clarifies why allocation temperature should be treated as an additional reservoir dimension rather than a renamed gain or sparsity parameter. Temperature changes continuous row concentration with relatively small spectral-radius variation. Gain strongly changes spectral radius while leaving row concentration fixed. Leak rate changes update timescale. Hard sparsity changes the zero/nonzero edge pattern. No single path dominates every task operationally, but their different structural signatures make them complementary controls.

The distribution-shift study broadens the question from sensitivity along one accepted temperature path to structural model choice over a two-dimensional temperature–leak bank. The target-label upper bound shows that substantial improvement is available when target errors are known. The pre-deployment sparse ensemble recovers roughly half of that reduction at the task-average level without using held-out labels. The unlabeled error-prediction model does not outperform that simpler ensemble, which shows that estimating target error from unlabeled target statistics remains difficult. The failure under block missingness is especially informative: a fixed structural ensemble cannot compensate for information that is absent from the input.

Table 10 summarizes conservative actions supported by the evidence. Low prediction variation favors retaining the validation-selected default because the accepted temperatures produce similar outputs. High variation is a monitoring signal, not an automatic direction of movement. If a shift type has been validated previously, a pre-deployment ensemble may provide a useful fallback. Information-removing shifts call for missing-data handling, abstention, fallback sensing, recalibration, or delayed-label intervention.

Table 10: Operational interpretation of the path-sensitivity indicators. The table separates observable prediction sensitivity from the unobserved target error of each structural candidate.

Observation	Defensible response before target labels are available
Low prediction variation across the safe region	Retain the validation-selected default because the accepted temperatures produce similar outputs.
High prediction variation, with no previously validated shift type	Flag structural sensitivity and continue with the default or another separately validated fallback until target information becomes available.
High variation under a previously validated noise or drift condition	A pre-deployment sparse ensemble can be considered if its behavior has already been validated for that condition.
Block missingness or another information-removing shift	Use missing-data handling, fallback sensing, abstention, or recalibration rather than relying on structural averaging alone.
Delayed targets become available	Re-estimate which structural candidates have low error and recalibrate the operating choice.

The validation-safe temperature region $\mathcal{T}_{\text{safe}}$ and the two-dimensional certified subset Θ_{cert} serve related but different purposes. The former is a connected interval around the validation-selected temperature. The latter is a pre-deployment filter over a temperature–leak grid. Restricting the target-label upper bound to Θ_{cert} increases mean loss by about 5%–10% across the three shift-study tasks. This quantifies the trade-off between conservative pre-deployment filtering and access to structures that become useful only under particular target shifts.

The retained historical real-world summaries are consistent with the same interpretation: Appliances Energy gives a null safe-region-versus-control result, whereas Air Quality gives a positive one. Because current public-data reconstructions do not numerically recover those operational

point estimates, we treat this panel as supporting historical evidence rather than independently reproduced validation.

12 Limitations

The results define a useful but bounded scope for the method.

1. The path-sensitivity indicators do not identify the temperature with the lowest target error. Without target labels, the validation-selected default remains the justified automatic choice unless a separate fallback policy has been validated.
2. The distribution-shift study shows that unlabeled prediction and state statistics are not sufficient, in the evaluated setting, to reliably rank all structural candidates by target error. The target-label upper bound is therefore an upper bound on available improvement rather than a deployable adaptation rule.
3. The pre-deployment sparse ensemble is shift-dependent. It improves performance under stronger correlated noise and offset drift, but degrades under block missingness. Information-removing shifts require additional mechanisms.
4. Allocation temperature is structurally distinct from gain, leak rate, and hard sparsity, but it does not dominate them on every task. Its main contribution is a new, interpretable structural coordinate and a calibration framework for studying that coordinate.
5. The evaluated reservoirs range from $N = 45$ to $N = 150$, with temperature grids of 13 or 25 points in the one-dimensional studies. Larger reservoirs, alternative raw-score distributions, sparse or modular topologies, physical reservoirs, and joint changes of grid density and reservoir size remain to be tested.
6. Some uncertainty summaries contain few independent higher-level units. In the distribution-shift study, only three independent reservoir substrates are available per task, so Table 8 reports the observed task means without emphasizing task-specific bootstrap intervals. The shift-type intervals in Table 9 resample nine task-substrate clusters and should still be interpreted as finite-panel uncertainty summaries.
7. The entropy result characterizes the recurrent allocation itself. It does not prove task optimality, echo-state guarantees, or universal improvements in memory capacity as temperature changes.

13 Conclusion

Allocation temperature adds a distinct structural configuration dimension to reservoir computing. It continuously changes how recurrent weight mass is distributed across incoming connections, while a separate normalization keeps bulk recurrent scale comparable. Row entropy and effective support increase monotonically with temperature, giving the parameter a direct structural interpretation that differs from recurrent gain, leak rate, and hard sparsity.

Validation turns this structural axis into an empirical operating path. The validation-safe temperature region is enriched for useful held-out temperatures relative to same-width control intervals, and prediction variation across that region indicates when temperature choice matters more on a test window. These conclusions persist under a denser temperature grid and a larger reservoir, although the direct NRMSE improvement available inside the safe region is modest and fixed binary thresholds are less robust than the continuous indicator relationship.

The final distribution-shift study shows why the structural dimension remains interesting beyond those modest within-region gains. When temperature and leak rate are considered jointly, target labels reveal much larger structural improvement than can be identified automatically from unlabeled target data. A sparse ensemble fitted before deployment recovers part of that opportunity, but its reliability depends on the type of shift and breaks down under block missingness. Allocation temperature therefore provides both a new reservoir design coordinate and a controlled way to study when recurrent allocation matters, what can be inferred without labels, and where additional adaptation mechanisms are still required.

Code and Data Availability

The public reproducibility package is available in the [project repository](#). It contains the manuscript source, frozen manuscript tables and figures, environment specifications, computational notebooks, and verification checks. Appendix C maps each main-text result to the corresponding computational artifact. The repository distinguishes original executed analyses retained for M23, SRP04b, and SRP05 from explicitly labelled publication-reconstruction notebooks for M19a–M21, whose original June 2026 notebook binaries could not be recovered. Historical manuscript values and fresh reconstructed outputs are kept separate, with the distinction documented in `PROVENANCE.md`. Fresh M20b/M20c2 executions recover the documented structural path but not the retained historical operational point estimates. The public datasets are the UCI Appliances Energy Prediction and UCI Air Quality datasets; they are not redistributed in the repository, and `data/README.md` provides the expected filenames and source information.

A Softmax allocation and entropy monotonicity

A.1 Softmax solution of the entropy-regularized row allocation

For completeness, we derive the row-wise softmax form used in Eq. (2). Fix a reservoir row i , write $a_j = a_{ij}$, and consider

$$p^\star = \arg \max_{p \in \Delta} \left[\sum_j p_j a_j + \tau H(p) \right], \quad H(p) = - \sum_j p_j \log p_j, \quad (21)$$

where $\Delta = \{p : p_j \geq 0, \sum_j p_j = 1\}$. Expanding the entropy term gives

$$\sum_j p_j a_j - \tau \sum_j p_j \log p_j. \quad (22)$$

Introducing a Lagrange multiplier λ for the normalization constraint, the Lagrangian is

$$\mathcal{L}(p, \lambda) = \sum_j p_j a_j - \tau \sum_j p_j \log p_j + \lambda \left(1 - \sum_j p_j \right). \quad (23)$$

At an interior optimum,

$$\frac{\partial \mathcal{L}}{\partial p_j} = a_j - \tau(\log p_j + 1) - \lambda = 0. \quad (24)$$

Therefore

$$\log p_j = \frac{a_j}{\tau} - 1 - \frac{\lambda}{\tau}, \quad p_j = C \exp(a_j/\tau), \quad (25)$$

where $C = \exp(-1 - \lambda/\tau)$ is independent of j . Enforcing $\sum_j p_j = 1$ gives

$$p_j = \frac{\exp(a_j/\tau)}{\sum_k \exp(a_k/\tau)}. \quad (26)$$

Restoring the row index gives Eq. (2).

A.2 Derivative of row entropy with respect to temperature

For a fixed row i , suppress the row index by writing $a_j = a_{ij}$ and $p_j(\tau) = p_{ij}(\tau)$. Then

$$p_j(\tau) = \frac{\exp(a_j/\tau)}{Z(\tau)}, \quad Z(\tau) = \sum_k \exp(a_k/\tau), \quad \mu(\tau) = \sum_j p_j(\tau) a_j = \mathbb{E}_{J \sim p(\tau)}[a_J]. \quad (27)$$

The expectation notation is only shorthand for a temperature-dependent weighted average over the fixed score list $\{a_j\}$. Equivalently, one may imagine drawing an index J from the categorical distribution $p(\tau)$ and then reading the corresponding fixed score a_J .

Since $\log p_j = a_j/\tau - \log Z(\tau)$ and $\sum_j p_j = 1$,

$$H(\tau) = - \sum_j p_j(\tau) \log p_j(\tau) \quad (28)$$

$$= - \sum_j p_j(\tau) \left(\frac{a_j}{\tau} - \log Z(\tau) \right) \quad (29)$$

$$= \log Z(\tau) - \frac{1}{\tau} \sum_j p_j(\tau) a_j \quad (30)$$

$$= \log Z(\tau) - \frac{\mu(\tau)}{\tau}. \quad (31)$$

This is Eq. (7) with the row index suppressed. Moreover,

$$\frac{d}{d\tau} \log Z(\tau) = -\frac{\mu(\tau)}{\tau^2}. \quad (32)$$

To differentiate μ , use $\beta = 1/\tau$. For the Gibbs distribution $p_j \propto \exp(\beta a_j)$,

$$\frac{d\mu}{d\beta} = \text{Var}_{p(\tau)}(a), \quad \frac{d\beta}{d\tau} = -\frac{1}{\tau^2}, \quad (33)$$

so

$$\frac{d\mu}{d\tau} = -\frac{\text{Var}_{p(\tau)}(a)}{\tau^2}. \quad (34)$$

Therefore,

$$\frac{dH}{d\tau} = -\frac{1}{\tau} \frac{d\mu}{d\tau} = \frac{\text{Var}_{p(\tau)}(a)}{\tau^3} \geq 0. \quad (35)$$

Restoring the row index proves Eq. (8). The sign is positive because increasing τ softens the allocation distribution, decreasing the allocation-weighted mean score $\mu_i(\tau)$ but increasing entropy. Finally, Eq. (9) inherits the same monotonicity because

$$\frac{dS_{\text{eff},i}}{d\tau} = S_{\text{eff},i}(\tau) \frac{dH_i}{d\tau} = S_{\text{eff},i}(\tau) \frac{\text{Var}_{p_i(\tau)}(a_i)}{\tau^3} \geq 0.$$

Thus the derivation explicitly connects the softmax allocation, the entropy identity, its monotonic derivative, and the monotonic effective-support statement.

B Secondary binary threshold analysis

The binary threshold analyses are retained for completeness but are not primary claims. They depend on the base positive-gain rate, the top-quartile threshold, and the number of distinct operating points inside each safe region. In the original path-comparison and real-data studies, high-indicator windows showed descriptive enrichment, as summarized in Table 11. Under the grid-density and reservoir-size interventions, however, the same binary precision ratio is near or below one and its confidence intervals cross one. The primary indicator result is therefore the continuous association between prediction variation and available safe-region improvement.

Table 11: Secondary top-quartile threshold analysis for the original path-comparison and real-data studies. High-indicator windows are the top quartile of safe-region dispersion or low/high endpoint spread. These values are descriptive for the corresponding study settings.

Study	Base positive-gain rate	High-indicator precision	Precision ratio	High-minus-low mean gain
Synthetic path comparison	0.40	0.65–0.67	1.62–1.68×	≈ 0.025
Appliances Energy	0.275	0.50–0.55	1.82–2.00×	≈ 0.0155
Air Quality	0.433	0.867	2.00×	0.087–0.104

Table 12: Secondary binary positive-gain precision ratio under the grid-density and reservoir-size interventions. Values close to or below one show that a fixed top-quartile threshold is less stable than the continuous indicator relationship.

Configuration	Safe-dispersion lift	Anchor-spread lift
$N = 60, K = 13$	1.026	0.999
$N = 60, K = 25$	0.934	0.955
$N = 150, K = 13$	0.925	0.991

C Reproducibility map

Table 13 provides the file-level mapping from manuscript results to the computational record. The public repository contains the publication notebooks, frozen manuscript tables, fresh reconstructed outputs where applicable, figure assets, environment specifications, and verification checks. Repository provenance is explicit: M23, SRP04b, and SRP05 retain their original executed analyses, whereas M19a–M21 are publication reconstructions because the original June 2026 notebook binaries could not be recovered.

Table 13: Mapping from manuscript results to computational notebooks. Notebook identifiers appear here only as reproducibility pointers.

Manuscript location	Main result	Notebook(s)
Section 5, Table 2, Figs. 2–4	Core transfer, matched controls, path-sensitivity indicators, standard-task boundary	M19a, M19b, M19c, M19d
Section 6, Table 3, Fig. 5	Comparison with gain, leak rate, and hard sparsity	M20a
Section 7, Table 4	Retained historical Appliances Energy and UCI Air Quality summaries	M20b, M20c2 (public-data reconstructions)
Section 8, Table 5, Appendix B	Safe-region width, tolerance, spectral-radius comparison, secondary threshold analysis	M21
Section 9, Tables 6–7, Fig. 6	Grid-density and reservoir-size robustness	M23
Section 10, Tables 8–9	Distribution-shift comparison and matched candidate-set upper bounds	SRP04b; SRP05 post-processing

Repository: <https://github.com/ildefons/temperature-calibrated-reservoirs>.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT to assist with language refinement, manuscript organization, clarification of the theoretical presentation, and the drafting and debugging of computational code. The author drafted the theoretical arguments and manuscript content, critically evaluated, revised, and verified all theory, computational code, experimental results, and manuscript material, made all final scientific decisions, and takes full responsibility for the published work.

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